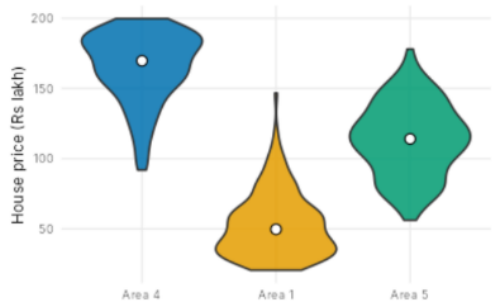
R.No: 38 Reg. No: 192425184 Name: U ROSHITHA

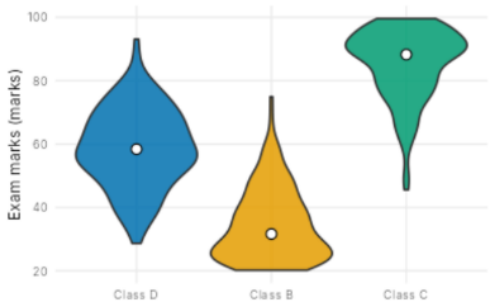
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: House price (Rs lakh)



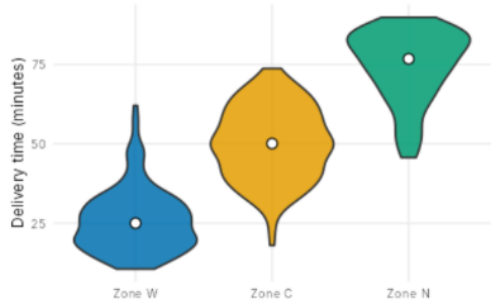
Violin	Skew (circle one)	Mean vs Median (circle)
Area 4	Right / Left / Symm.	Mean > / < / ~ Median
Area 1	Right / Left / Symm.	Mean > / < / ~ Median
Area 5	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Exam marks (marks)



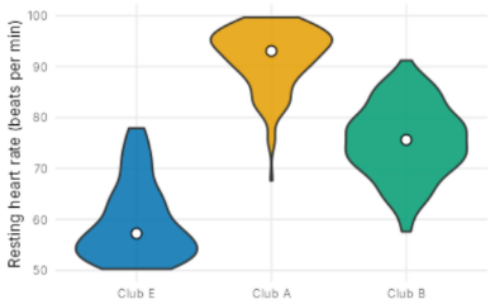
Violin	Skew (circle one)	Mean vs Median (circle)
Class D	Right / Left / Symm.	Mean > / < / ~ Median
Class B	Right / Left / Symm.	Mean > / < / ~ Median
Class C	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Delivery time (minutes)



Violin	Skew (circle one)	Mean vs Median (circle)
Zone W	Right / Left / Symm.	Mean > / < / ~ Median
Zone C	Right / Left / Symm.	Mean > / < / ~ Median
Zone N	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Resting heart rate (beats per min)



Violin	Skew (circle one)	Mean vs Median (circle)
Club E	Right / Left / Symm.	Mean > / < / ~ Median
Club A	Right / Left / Symm.	Mean > / < / ~ Median
Club B	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 2, violin "Class B" is not symmetric. Towards which side is its tail, and which is the better measure of a typical exam marks here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Activity: Violin Plot Analysis Using R and ggplot2

Part A – Analysis of Violin Plots

Chart 1: House Price

Violin	Skew	Mean vs Median
Area 4	Left	Mean < Median
Area 1	Right	Mean > Median
Area 5	Right	Mean > Median

Chart 2: Exam Marks

Violin	Skew	Mean vs Median
Class A	Right	Mean > Median
Class B	Right	Mean > Median
Class C	Left	Mean < Median

Chart 3: Delivery Time

Violin	Skew	Mean vs Median
Zone W	Right	Mean > Median
Zone C	Symmetric	Mean ≈ Median
Zone N	Left	Mean < Median

Chart 4: Resting Heart Rate

Violin	Skew	Mean vs Median
Club E	Right	Mean > Median
Club A	Left	Mean < Median
Club B	Symmetric	Mean ≈ Median

Part B

B1. Answer

Class B has a right-skewed distribution because its tail extends towards higher marks. The median is the better measure of typical exam marks because it is less affected by skewed values.

Tail: Right side
Better measure: Median

B2. R Code

```
ggplot(df, aes(x = group, y = value)) +
  geom_violin() +
  stat_summary(fun = median, geom = "point", color = "white")
```

Conclusion

Violin plots help us understand the distribution and skewness of data. A right-skewed distribution generally has **Mean > Median**, while a left-skewed distribution has **Mean < Median**. For skewed data, the **median** is usually a better measure of the typical value.